

Aheadd CP-1 NXG Specimen Build

Statement of Work (SOW)

The National Aeronautics and Space Administration (NASA), specifically Marshall Space Flight Center (MSFC) has a need for manufacturing of Aheadd CP-1 material testing and characterization specimens using a Nikon SLM Solutions NXG additive manufacturing machine to evaluate the viability of using this platform for cryogenic fluid management (CFM) components and to better characterize material produced on large scale laser powder bed fusion machines.

The SOW is as follows:

1. The vendor shall have experience manufacturing Aheadd CP-1 with laser powder bed fusion (L-PBF) additive manufacturing.
2. The vendor shall develop a build layout using vendor's best practices, the specimen matrix, and build files or geometries provided by NASA, and review with NASA prior to manufacturing.
3. The provided build files shall be processed by the vendor using best practices to ensure the highest resolution feasible for the specimens and that sufficient stock material is added to account for build plate removal.
4. The vendor shall print labels on all specimens notating the build and specimen number.
5. The vendor shall de-powder the specimens and remove from the build plate using wire EDM or bandsaw in the as-built condition.
6. NASA will provide detailed specimen matrix, specimen files, and discuss as necessary.

Specimen Matrix

Part Name	Contents		
	QTY/Type	Total Printed Parts	File/Geometry
Tensile Specimens	40 Vertical	43	Tensile_ASTM_E8_Short
	18 Horizontal (3 stacks of 6)		Stacked TN_1
HCF	30 Vertical	45	HCF Prepolish
	12 Vertical As-Built		HCF As-Built
	18 Horizontal (3 stacks of 6)		Stacked TN_1
LCF	48 Vertical	48	LCF Polish
Metallography and Thermophysical	20 Blank Cylinders	20	0.6" diameter 4" tall
Thin sections	8 grouped specimens	8	Thin Section with Stock

Laser Quality	12	12	Nikon recommended geometry
Challenge Parts	Small geometric parts.	4	2 lattice boxes with shroud 2 lattice boxes without a shroud
Powder Cups	2	2	Nikon recommended geometry
Cryo CTE	4 Rectangular Blocks	4	1.25"x0.6"x4" block
Fatigue Crack Growth	6 Blocks	6	3"x0.75"x3"

Shipping

- Contractor shall ship all deliverables and returned samples to:
Marshall Space Flight Center Central Receiving,
Huntsville, AL 35812

Deliverables

- Printed Ahead CP-1 specimens per the specimen matrix provided and agreed upon.
- Documentation on the build pedigree including but not limited to:
 - Powder certificate of conformance with material composition for powder lots used
 - Laser assignment and parameters that may be included in a standard job report.

Period of Performance

The tasks shall be performed within ten (10) weeks of contract award.